

Attractor Networks, Spatial Maps, and Memory Consolidation

From Hopfield Networks to Grid Cells and Complementary Learning Systems

Outline

Part 1: Hopfield Networks and Associative Memory

- Energy function, Hebbian learning, storage capacity
- Statistical mechanics and phase transitions
- Modern Hopfield networks and transformer attention

Part 2: Grid Cell Attractor Models

- Continuous attractor networks and bump dynamics
- Ring attractor $\rightarrow$ torus $\rightarrow$ twisted torus $\rightarrow$ hexagonal grids
- Burak & Fiete (2009): velocity integration and path integration

Part 3: Memory Consolidation

- Complementary learning systems: hippocampus vs. neocortex
- Replay, sharp-wave ripples, and sleep consolidation

Part 1

Hopfield Networks and Associative Memory

Associative memory

Content-addressable memory: retrieve a stored pattern from a partial or noisy cue

- Store P binary patterns, each of N bits
- Given a corrupted version of pattern μ , recover the full pattern
- Unlike a lookup table — the address is the content itself
- The brain does this: a partial face, a few notes of a melody, a fragment of a sentence
→ full memory

Hopfield's insight (1982): a recurrent neural network whose attractors are the stored memories

The Hopfield network

N binary neurons with symmetric, all-to-all connectivity:

$$x_i \in [-1, +1], \quad i = 1, \dots, N$$

Update rule (asynchronous):

$$x_i(t+1) = \operatorname{sgn} \left(\sum_j W_{ij} x_j(t) \right)$$

- Pick a random neuron i , compute its local field, update its state
- Repeat until no neuron changes — the network has converged
- Symmetric weights ($W_{ij} = W_{ji}$) guarantee convergence

The energy function

Every Hopfield network has a Lyapunov function that decreases with each update:

$$E = -\frac{1}{2} \sum_{i \neq j} W_{ij} x_i x_j$$

Key property:

$$\Delta E = - \Delta x_i \sum_j W_{ij} x_j \leq 0$$

- Energy can only decrease or stay the same — never increase
- The network always converges to a local minimum of E
- Local minima = fixed-point attractors = stored memories
- Dynamics = gradient descent on the energy landscape

The Hebbian learning rule

Store P patterns as fixed-point attractors using the outer product rule:

Stored patterns:

$$\xi^1, \xi^2, \dots, \xi^P \in [-1, +1]^N$$

Weight matrix:

$$\mathbf{W} = \frac{1}{N} \sum_{\mu=1}^P \xi^\mu (\xi^\mu)^\top \quad W_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^\mu \xi_j^\mu$$

- "Neurons that fire together, wire together" — sum of outer products
- Each pattern contributes additively to the weight matrix
- One-shot, local, Hebbian — biologically plausible

Pattern retrieval

Initialize the network with a noisy cue, let it converge, measure the overlap:

Overlap (order parameter):

$$m^\mu = \frac{1}{N} \sum_i \xi_i^\mu x_i$$

$$m^\mu \approx 1 \quad \Longleftrightarrow \quad \mathbf{x} \approx \boldsymbol{\xi}^\mu$$

Signal-to-noise analysis:

When the network is in state $\mathbf{x} \approx \boldsymbol{\xi}^\mu$, the local field at neuron i is:

$$h_i = \underbrace{\xi_i^\mu}_{\text{signal}} + \underbrace{\frac{1}{N} \sum_{\nu \neq \mu} \xi_i^\nu \sum_j \xi_j^\nu \xi_j^\mu}_{\text{cross-talk noise}}$$

Storage capacity

How many patterns can the network reliably store and retrieve?

$$P_{\max} = \alpha_c N, \quad \alpha_c \approx 0.138$$

1. Cross-talk noise scales as P/N

Each additional pattern adds interference — when the noise overwhelms the signal, retrieval fails

2. Phase transition at $\alpha \approx 0.138$

Below this ratio: perfect retrieval. Above: catastrophic forgetting — all patterns lost simultaneously

3. Example: $N = 1000$ neurons

Can store ~ 138 patterns. $N = 10,000 \rightarrow \sim 1,380$ patterns

4. Sharp transition, not gradual

Derived exactly by Amit, Gutfreund & Sompolinsky (1985) using replica theory from spin glass physics

Statistical mechanics of memory

Amit, Gutfreund & Sompolinsky (1985, 1987)

The Hopfield network is a spin glass:

- Neurons $x_i \in \{-1, +1\} \leftrightarrow$ Ising spins
- Symmetric weights $W_{ij} \leftrightarrow$ coupling constants J_{ij}
- Energy $E = -\frac{1}{2} \sum W_{ij} x_i x_j \leftrightarrow$ Hamiltonian H
- Asynchronous dynamics $\leftrightarrow$ single spin-flip Glauber dynamics
- Stored patterns $\xi^u \leftrightarrow$ quenched disorder

Phase diagram (as a function of $\alpha = P/N$ and temperature T):

$\alpha < 0.138$, low T

Retrieval phase — patterns are stable attractors

$\alpha > 0.138$

Spin glass phase — too many patterns, catastrophic failure

High T

Paramagnetic phase — thermal noise destroys all attractors

Limitations of classical Hopfield networks

1. Low capacity

Only $\sim 0.14N$ patterns in N neurons — far less than the $N^2/2$ available synapses could support

2. Spurious attractors

Mixture states (linear combinations of stored patterns) and spin-glass states are also stable — false memories

3. Correlated patterns

The outer product rule fails for correlated patterns — cross-talk becomes systematic, not random

4. Binary neurons only

The original model uses ± 1 neurons — real neurons have graded, continuous firing rates

5. No temporal sequences

Only stores static patterns, not sequences — requires extensions (e.g., asymmetric weights)

Modern Hopfield networks

Ramsauer et al. (2020); Krotov & Hopfield (2016)

Replace the quadratic energy with an exponential energy:

$$E = -\log \sum_{\mu} \exp(\xi^{\mu} \cdot \mathbf{x})$$

Update rule:

$$\mathbf{x}_{\text{new}} = \sum_{\mu} \text{softmax}(\beta \xi^{\mu} \cdot \mathbf{x}) \xi^{\mu}$$

Key advantage — exponential storage capacity:

$$P_{\text{max}} \sim \exp(\alpha N)$$

Connection to transformer attention

The modern Hopfield update is equivalent to one step of self-attention:

Transformer attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}$$

Correspondence:

Hopfield	Transformer	Role
Query state x	Query Q	Current state / probe
Stored patterns ξ^u	Keys K	Memory entries
Stored patterns ξ^u	Values V	Retrieved content
$\exp(\xi^u \cdot x)$	$\text{softmax}(QK^\top/\sqrt{d})$	Similarity $\rightarrow$ weight
1 retrieval step	1 attention head	One pattern completion

Part 1 Summary: Hopfield Networks

1. The Hopfield network stores memories as fixed-point attractors of a recurrent network
2. Energy $E = -\frac{1}{2} \mathbf{x}^T \mathbf{W} \mathbf{x}$ guarantees convergence to local minima
3. Hebbian outer product rule: one-shot, local, biologically plausible
4. Storage capacity $\alpha_c \approx 0.138 N$ — sharp phase transition (spin glass physics)
5. Modern Hopfield: exponential energy $\rightarrow$ exponential capacity $\rightarrow$ transformer attention

Key theme: attractor dynamics as a computational principle for memory

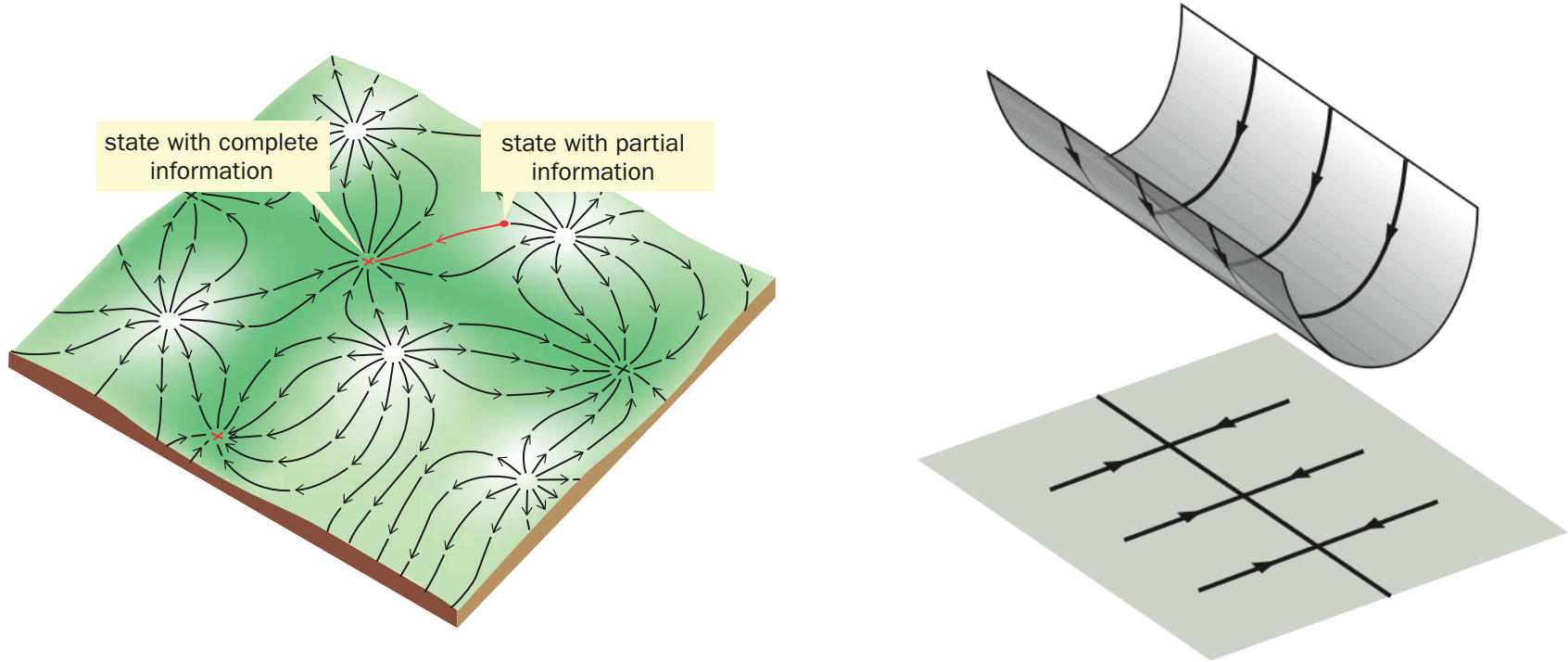
$\rightarrow$ Next: the same principle applied to spatial representation and navigation

Part 2

Grid Cell Attractor Models

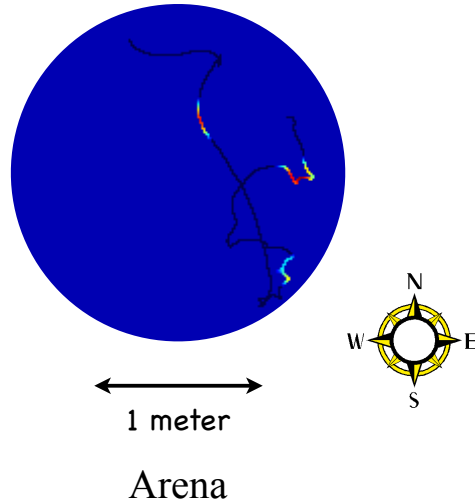
Continuous attractor networks, ring attractors, and the
Burak & Fiete (2009) model of path integration

From discrete to continuous attractors



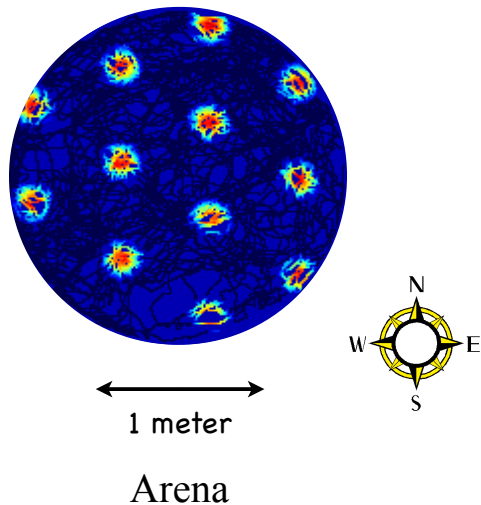
The grid cell system and path integral

Single neuron activity



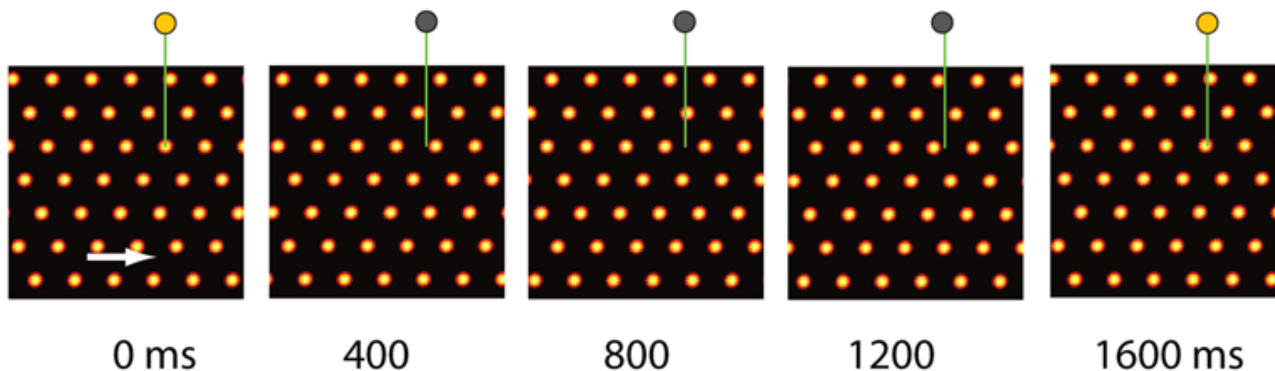
The grid cell system and path integral

Single neuron activity

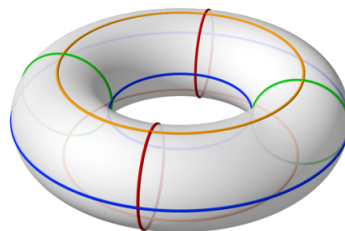


The grid cell system and path integral

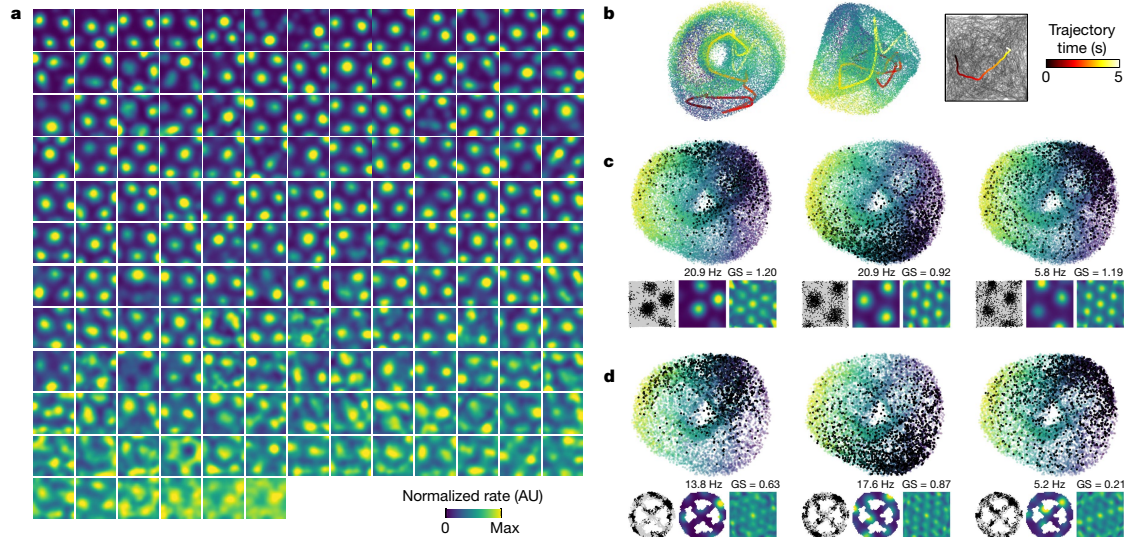
Population neural activity on a 2D neural sheet (illustration)



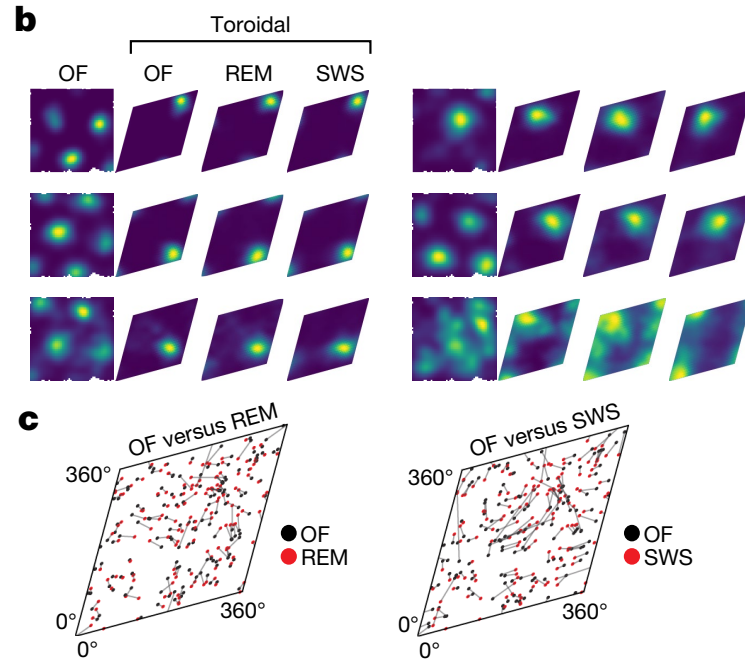
$$S^1 \times S^1$$



Toroidal structure of population grid cell activity

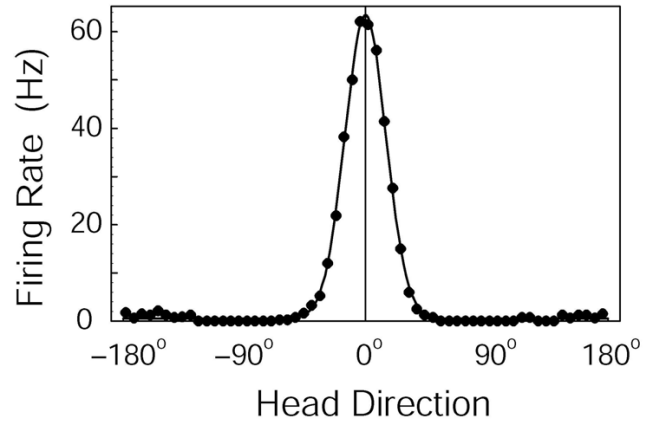


Toroidal structure persists during sleep (**rigid code!**)

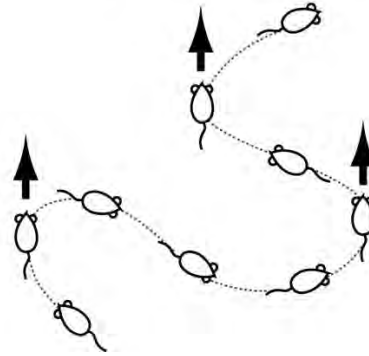


Head direction cell

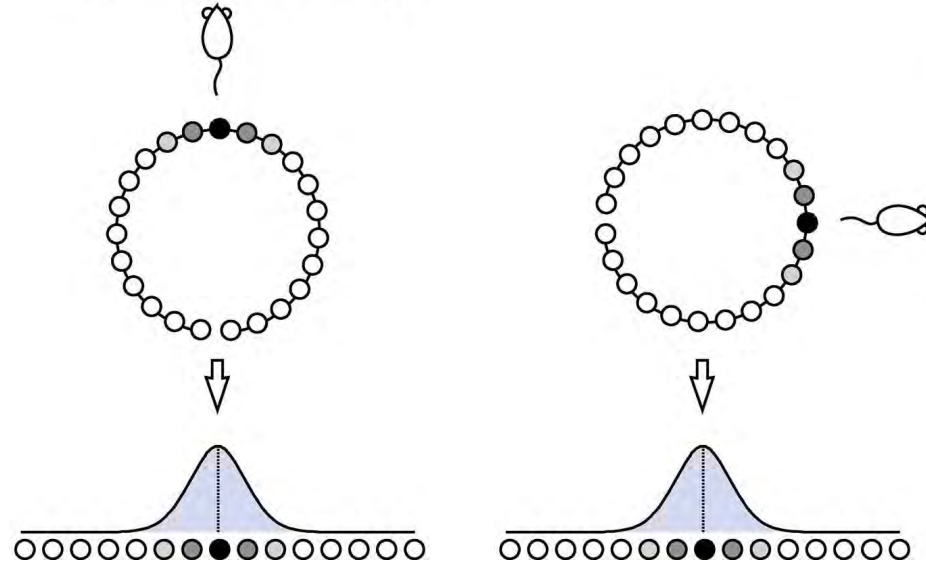
Tuning Curve of a Head-Direction Cell



Preferred Direction of a Single Head-Direction Cell



A ring attractor model

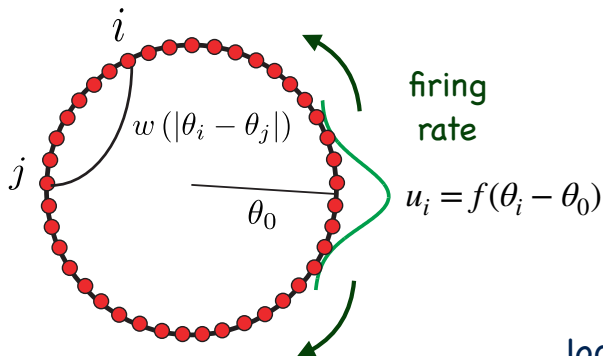


Ben-Yishai, Bar-Or, Sompolinsky, PNAS (1995)

Zhang, J. Neurosci. (1998)

Xie, Hahnloser, Seung, Phys. Rev. E (2002)

A ring attractor model



Solution:

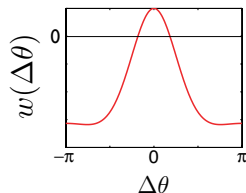
A continuum of steady states
encode a continuous parameter (angle)

e.g., $u(\theta) = a + b \cos(\theta - \theta_0)$

local excitation, long range inhibition

$$\tau \frac{du_i}{dt} = -u_i + \left[\sum_{j=1 \dots N} w_{ij} u_j + I_i^0 \right]_+$$

$$w_{ij} = w_{ji} = \frac{1}{N} J(|\theta_i - \theta_j|)$$

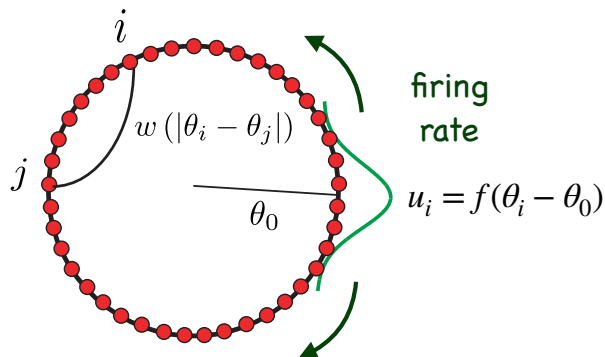


$$J = J_0 + J_1 \cos \theta$$

Let $N \rightarrow \infty$

$$\tau \frac{\partial u(\theta, t)}{\partial t} = -u(\theta, t) + \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} J(\theta - \theta') u(\theta', t) d\theta' + I^0(\theta) \right]_+$$

A ring attractor model



$$\tau \frac{du_i}{dt} = -u_i + \left[\sum_{j=1..N} w_{ij} u_j + I_i^0 \right]_+$$

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Let $N \rightarrow \infty$

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Solution:

A continuum of steady states
encode a continuous parameter (angle)

e.g., $u(\theta) = a + b \cos(\theta - \theta_0)$

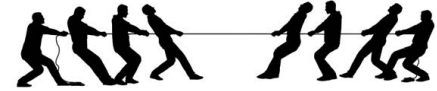
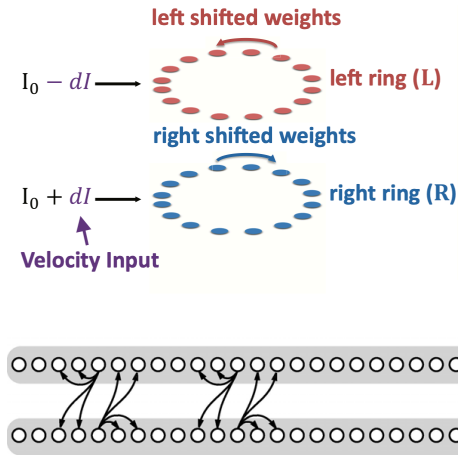
$$J = J_0 + J_1 \cos \theta + \gamma \sin \theta$$

Asymmetric term

The bump moves at a speed γ/τ

The double ring model

Synaptic input to neuron i :

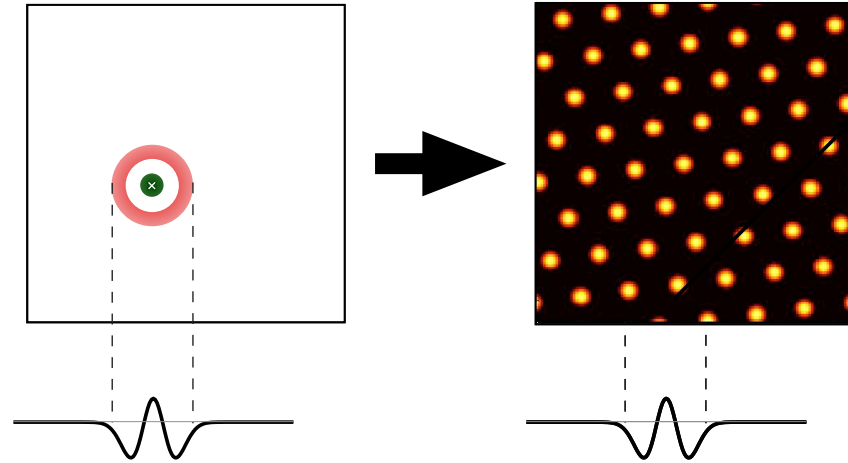
$$\sum_{j=1}^{2N} W_{ij} s_j + I_0 \mp dI$$


Two competing groups of cells try to pull the activity bumps in opposite directions.

Without any external input, the two groups are balanced and the activity bumps are aligned and stationary.

When the input (same for every neuron) to one group is stronger than the other, the balanced is altered and the two bumps both move in the same direction.

The grid attractor model



“Mexican hat” connectivity:

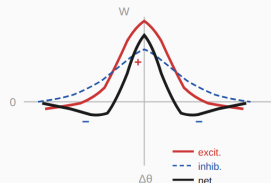
Local excitation,
longer range inhibition

$$w_{ij} = J(|\vec{r}_i - \vec{r}_j|) = J(\vec{r})$$

W is a Toeplitz matrix

From ring attractors to hexagonal grids

1D Ring Attractor



Center-surround connectivity
(difference of Gaussians)

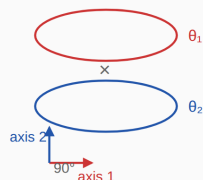
$$W(\Delta\theta) = A_e e^{-\Delta\theta^2/2\sigma_e^2} - A_i e^{-\Delta\theta^2/2\sigma_i^2}$$

Connectivity (Burak & Fiete)

$$W(\Delta\mathbf{r}) = A_e e^{-|\Delta\mathbf{r}|^2/2\sigma_e^2} - A_i e^{-|\Delta\mathbf{r}|^2/2\sigma_i^2}$$

Circulant 2D connectivity: narrow excitation, broad inhibition (DoG). Eigenmodes are Fourier modes on the torus. $\sigma_e < \sigma_i$.

Two Independent Rings

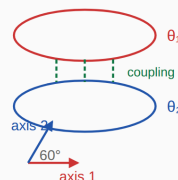


$S^1 \times S^1$ = standard torus
Axes at 90°

separable: $W_x \otimes W_y$

→ Square grid

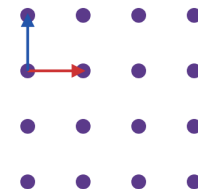
Two Coupled Rings



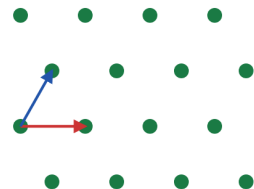
Twisted torus
Axes at 60° → coupling

$$\Delta x^2 + \Delta x \Delta y + \Delta y^2$$

→ Hexagonal grid



90° → Square lattice



60° → Hexagonal lattice

Separability → coupling

At 90°, each Gaussian in the DoG is separable:

$$e^{-|\Delta\mathbf{r}|^2/2\sigma^2} = e^{-\Delta x^2/2\sigma^2} \cdot e^{-\Delta y^2/2\sigma^2}$$

At 60° (oblique), cross-term couples the rings:

$$\Delta x^2 + \Delta y^2 = \Delta u^2 + \Delta u \Delta v + \Delta v^2$$

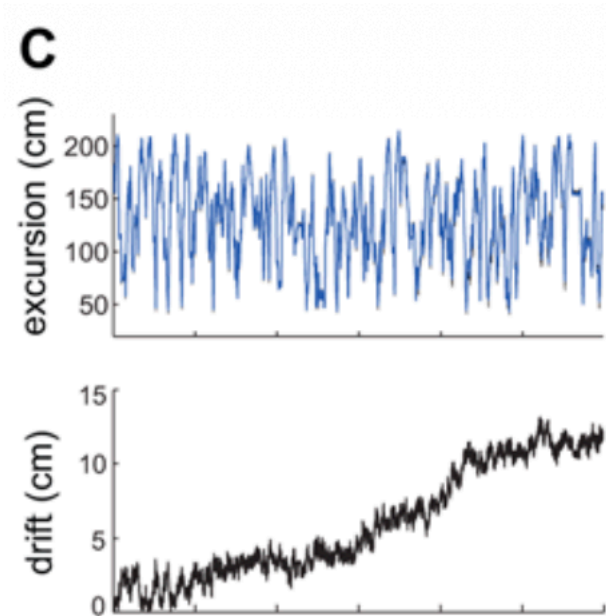
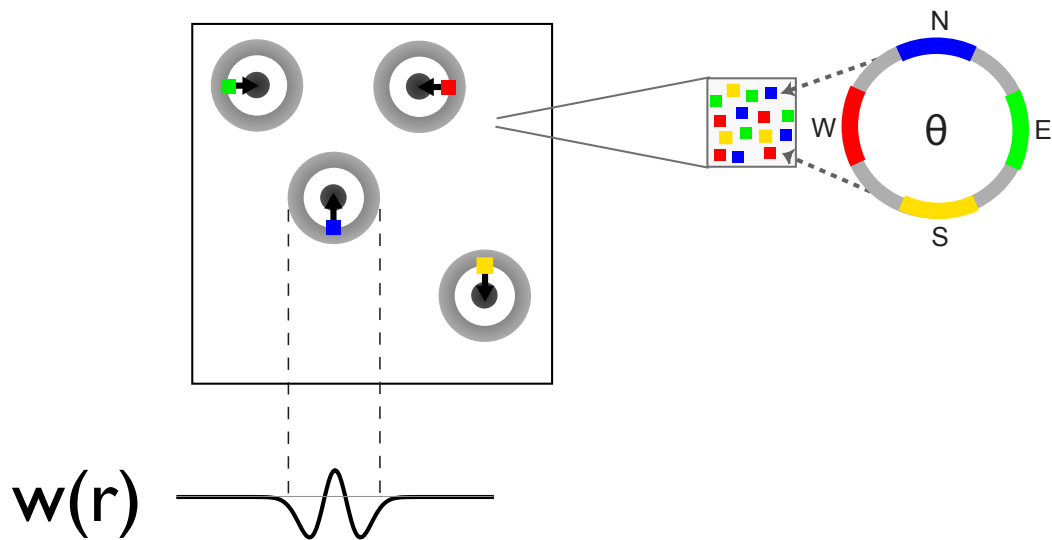
Fourier: why 2 rings suffice

Hexagonal pattern = 3 pairs of wavevectors, but only 2 independent because:

$$\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = 0$$

Each ring ↔ one conjugate pair. Two rings generate the full hexagonal symmetry.

How to perform path integration?



introduce asymmetry $w_{ij} = J(|\vec{r}_j + \gamma \vec{e}_{\theta_j} - \vec{r}_i|)$

external input $I_i = I_0 + \alpha \vec{v} \cdot \vec{e}_{\theta_i}$ $\vec{v}$: actual velocity of the animal

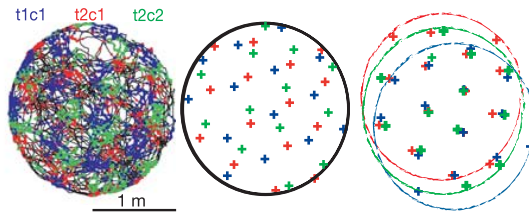
Burak & Fiete PloS Comp Biology 2009

Prediction of the attractor model

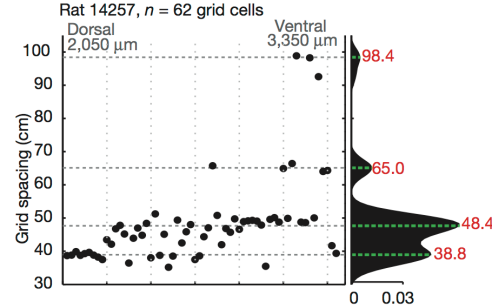
1. Modularity



2. Strict preservation of pairwise correlation structure
3. Topographic organization?



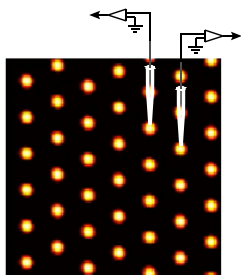
Hafting et al, *Nature* (2005)



Stensola et al, *Nature* 2012

Prediction of the attractor model

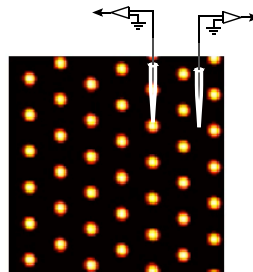
1. Modularity ✓
2. Strict preservation of pairwise correlation structure ✓
3. Topographic organization?



some neurons should always
fire together

Verified under: pharmacological,
genetic, optogenetic
perturbations

Yoon, Fiete, 2013, Review: Burak 2014



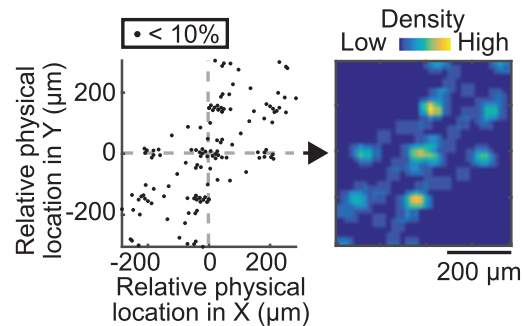
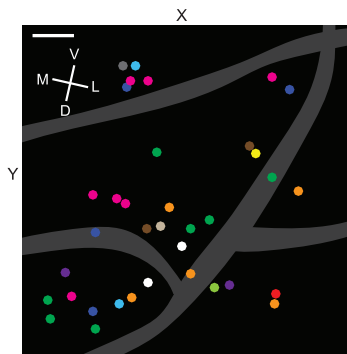
others should never
fire together

in sleep

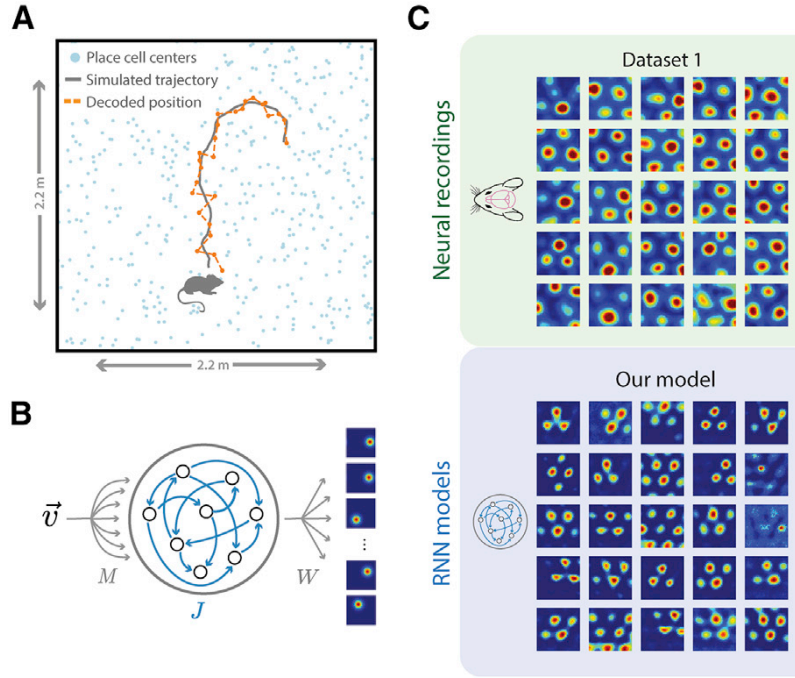
Gardner, ..., Moser 2019
Trettel, ..., Colgin 2019

Prediction of the attractor model

1. Modularity ✓
2. Strict preservation of pairwise correlation structure ✓
3. Topographic organization ✓ ?



Emergence of grid structure from a trained RNN

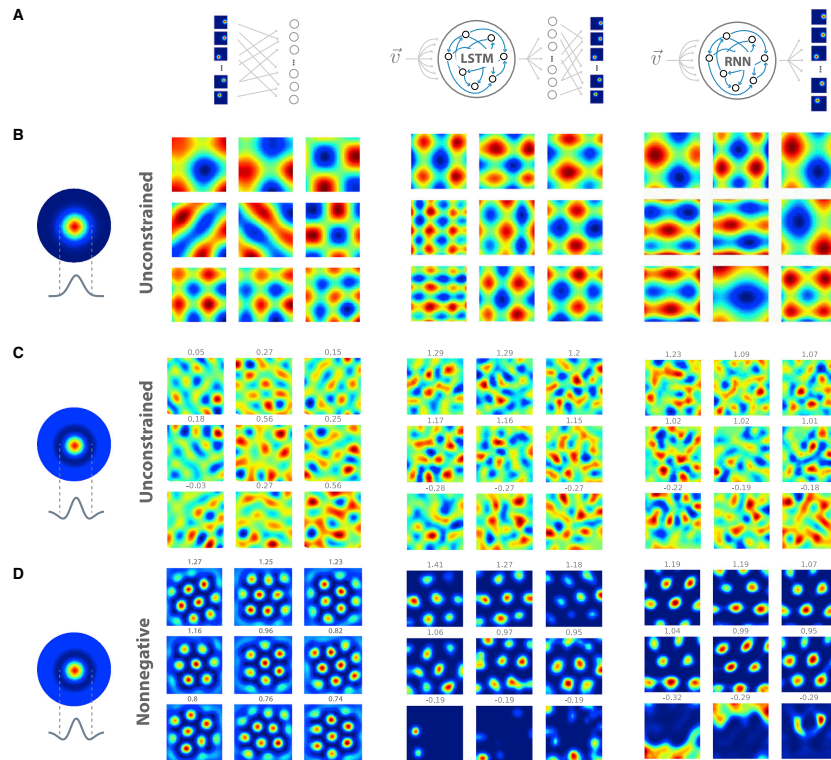


$$r_i^{t+1} = \sigma \left[\sum_{j=1}^{n_g} J_{ij} r_j^t + M_{ix} v_x^t + M_{iy} v_y^t \right]$$

$$\hat{p}_i^t = \sum_{j=1}^{n_g} w_{ij} r_j^t$$

Place cell Grid cell

Emergence of grid structure depends on the shape of place fields and other constraints



Part 3

Memory Consolidation

Complementary learning systems, hippocampal replay,
and the two-stage model of memory

One network cannot do both well

Memorize

- Encode a single new episode immediately
- Keep it distinct — do not overwrite old memories
- Sparse, non-overlapping codes
- Favours a high learning rate

Generalize

- Extract structure shared across many experiences
- Overlapping, distributed representations
- Average over examples — semantic knowledge
- Requires a low learning rate

These two demands pull a single set of weights in opposite directions.

Catastrophic interference

- Train one network on task B by gradient descent: weights move to minimize the new loss, degrading what task A had learned.
- New learning does not add to memory — it overwrites it (McCloskey & Cohen, 1989).

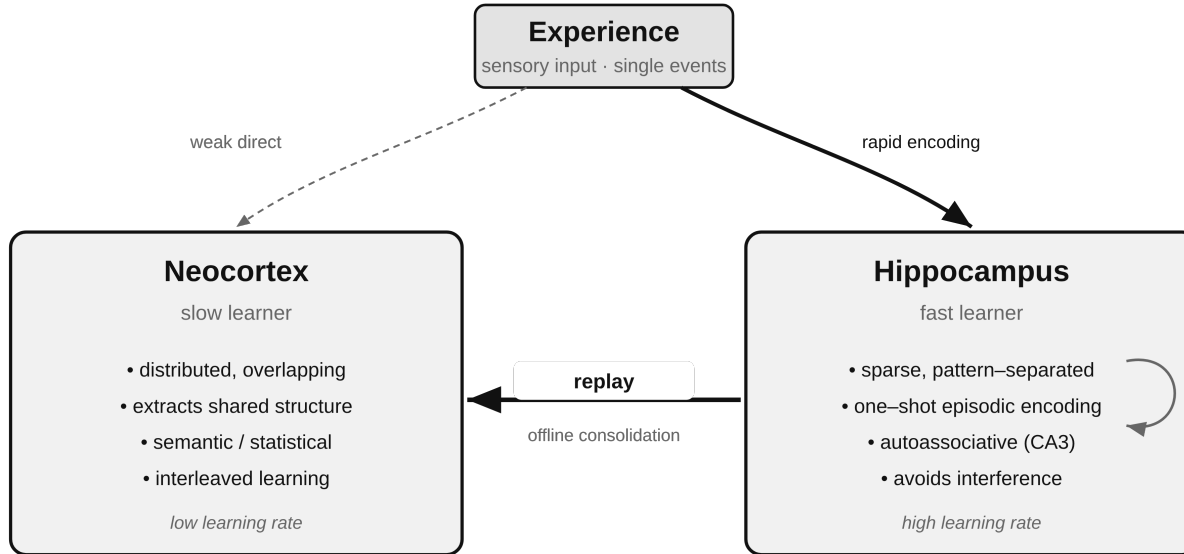
$$\theta_B^* = \arg \min_{\theta} \mathcal{L}_B(\theta) \implies \mathcal{L}_A(\theta_B^*) \uparrow$$

Same lesson from Part 1:

overload a Hopfield net and stored patterns blur into spurious minima — capacity is bounded.

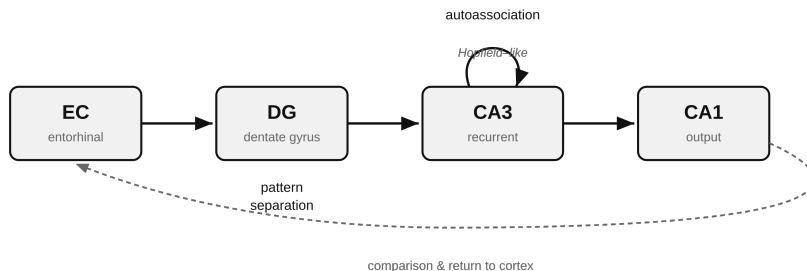
$$p_{\max} \approx 0.138 N$$

The solution: two complementary systems



McClelland, McNaughton & O'Reilly (1995)

The hippocampus is a fast attractor memory



- Dentate gyrus orthogonalizes inputs — pattern separation minimizes interference between episodes.
- CA3 recurrent collaterals form an autoassociative (Hopfield-like) net: a partial cue completes to the stored pattern.

$$\mathbf{x}_{\text{cue}} \longrightarrow \xi^{\mu}$$

pattern completion

Why cortex must learn slowly

The two systems run at very different learning rates:

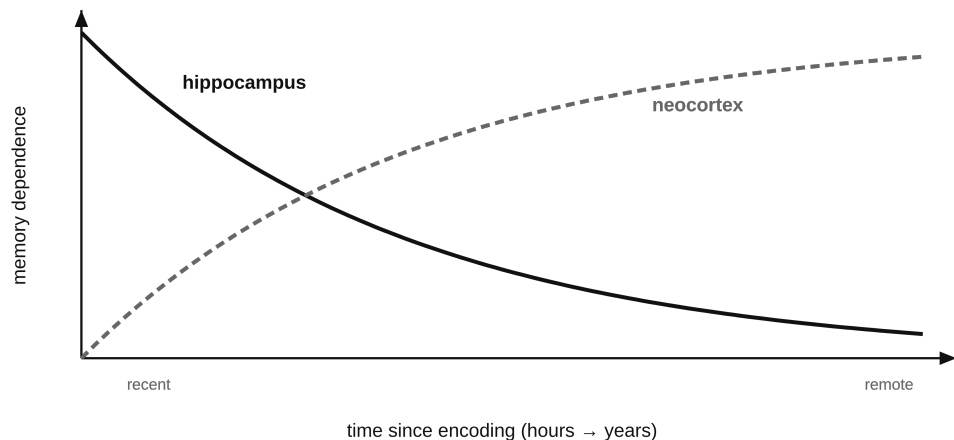
$$\alpha_{\text{hpc}} \gg \alpha_{\text{ctx}}$$

- Cortex discovers shared structure only by averaging over many interleaved examples:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim A \cup B} [\mathcal{L}(\theta; x, y)]$$

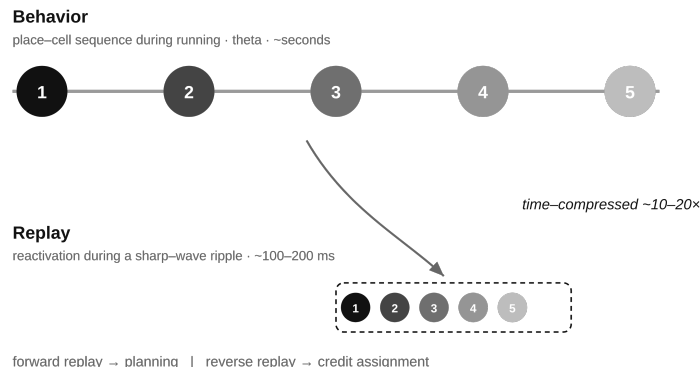
- Present examples sequentially instead and you get catastrophic interference.
- Replay supplies the interleaving offline — the hippocampus re-presents old experiences so cortex can integrate the new without erasing the old.

Systems consolidation



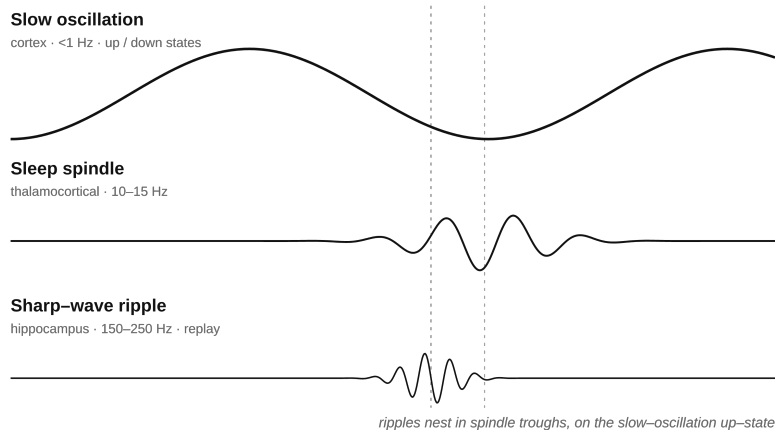
- Memories are initially hippocampus-dependent; over time they become supported by distributed cortical networks.
- Retrograde amnesia is temporally graded — recent memories are most vulnerable (Ribot gradient).
- Standard model: the trace transfers to cortex. Multiple-trace / transformation views: episodic detail stays hippocampal, gist becomes cortical.

The engine: hippocampal replay



- During rest and sleep, place-cell sequences reactivate inside sharp-wave ripples, time-compressed ~10–20x.
- Reverse replay supports credit assignment; forward replay supports planning. Replay is the teaching signal sent to cortex.

Sleep orchestrates the dialogue



- A triple nesting coordinates transfer: cortical slow oscillations gate thalamocortical spindles, which gate hippocampal ripples.
- Information flows hippocampus → cortex on the up-state. Disrupting ripples impairs consolidation (Girardeau et al., 2009).

Two levels of consolidation

Synaptic (cellular)

- Minutes to hours
- LTP stabilization, protein synthesis
- Local to the synapses just engaged
- Fixes the trace in place

Systems

- Days to years
- Reorganization across brain regions
- Hippocampus → neocortex transfer
- Driven by replay during sleep

Synaptic homeostasis (Tononi & Cirelli): sleep also renormalizes overall synaptic strength, improving signal-to-noise.

Same machinery, two jobs

- Replay is not only backward-looking consolidation. Forward replay simulates trajectories for planning — exactly the role of replay in Dyna and model-based RL.
- Mattar & Daw (2018): the brain prioritizes which experiences to replay by their expected value of a backup —

$$\text{EVB}(s, a) = \text{Gain}(s, a) \times \text{Need}(s, a)$$

Gain = how much replaying changes the policy. Need = how often the state will be visited.

One replay mechanism serves both memory consolidation and decision-making.

From brains to machines: experience replay

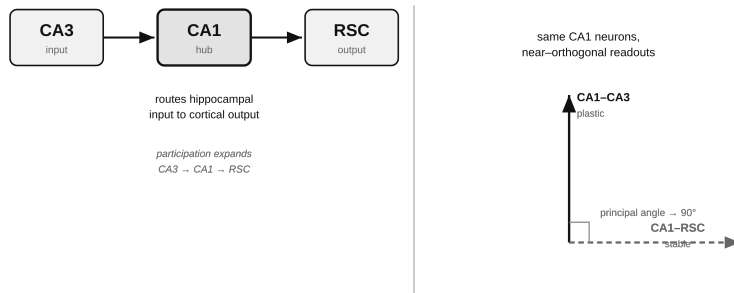
- Deep Q-Networks store transitions in a replay buffer and train on random minibatches drawn from it (Mnih et al., 2015):

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim U(\mathcal{D})} [(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta))^2]$$

- Sampling U(D) decorrelates consecutive experiences, reuses data, and stabilizes learning — directly inspired by hippocampal replay.
- Generative replay (Shin et al., 2017) replays synthesized past data to prevent catastrophic forgetting in deep nets — a CLS solution in machine learning.

One of the cleanest transfers from systems neuroscience to deep learning.

Communication subspaces in the hippocampal-cortical axis



- Up to 1,024-channel recordings across DG, CA3, CA2, CA1 and retrosplenial cortex (RSC); partial CCA extracts the subspaces these regions communicate through, with CA1 as the hub.
- The same CA1 neurons are recombined: input (CA1–CA3) and output (CA1–RSC) subspaces are rotated near-orthogonal, so one pool serves distinct routes without interference.
- An RNN trained to map CA3 → RSC discovered the same rotation; disabling it impaired learning — brains and ANNs converge on the trick.

Plasticity and stability, split by subspace

- **Replay is selective.** In post-experience NREM sleep — especially during sharp-wave ripples — the intrahippocampal CA1–CA3 subspace reactivated and reconfigured, while the hippocampo-cortical CA1–RSC subspace stayed stable.
- **A circuit-level division of labour.** CA3–CA1 prioritizes fast, flexible encoding; CA1–RSC preserves a stable long-term cortical readout — the two-stage CLS model, now localized to specific subspaces.
- **A concrete answer to catastrophic interference.** Rotating the CA3-coupled subspace orthogonally to the RSC-coupled one lets CA1 learn the new without disturbing what cortex reads out.
- **Time-division multiplexing.** Wake states encode through fast hippocampal plasticity; sleep states drive gradual cortical change — the wake / NREM split made mechanistic.

Complementary learning systems, observed at the level of single circuits — and a biological template for the orthogonal-readout tricks ML uses against forgetting.

Synthesis

- Hopfield networks — discrete attractors store memories; recall is pattern completion toward an energy minimum.
- Grid cells — continuous attractors carry a bump on a torus to represent and integrate space.
- Consolidation — a fast, sparse hippocampal attractor memory teaches a slow, distributed cortical learner; replay during sleep is the bridge.

Attractor dynamics + two learning timescales + replay = robust, interference-free lifelong learning — and the same principles reappear in modern AI.

Open questions

- **Transfer or transformation?** Does the memory relocate to cortex (standard model), or does the hippocampus stay permanently necessary for episodic detail while cortex keeps only the gist?
- **Does cortex really learn slowly?** Silent engrams form in prefrontal cortex on day one (Kitamura et al., 2017); a matching schema lets cortex consolidate within hours (Tse et al., 2007).
- **What selects replay — and is it generative?** How the brain prioritizes what to replay is unknown, and replayed shortcuts suggest the hippocampus constructs trajectories rather than playing back stored ones.
- **Active consolidation vs. synaptic homeostasis.** Selective strengthening and global synaptic downscaling make opposite predictions about what sleep does to synaptic weights.
- **The missing mechanistic bridge.** How a ~100 ms ripple instructs durable cortical plasticity, and what trace persists across years, is still largely a black box.

Why hasn't this solved continual learning in machines?

Experience replay and generative replay reduce catastrophic forgetting, but fall short of biological robustness. The gaps reveal what CLS actually requires:

- **Replay is a stored buffer, not a generative memory.** DQN stores raw transitions; the buffer cannot scale to a lifelong stream, and generative replay's own generator forgets.
- **No pattern separation.** Deep nets lack the dentate-gyrus orthogonalization that lets new episodes be stored without overwriting old ones.
- **No genuine two-timescale architecture.** A buffer bolted onto one network is a training trick — not a fast sparse memory teaching a slow distributed learner.
- **Unknown task boundaries.** ML methods usually assume known task boundaries; the brain faces an unsegmented stream and decides for itself when to consolidate.

Closing these gaps is as much a neuroscience question as an engineering one.

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